1 Symmetric Bilinear Forms

Definition 1.1 (Symmetric bilinear form)**.** A bilinear form $B \in V^{(2)}$ is called *symmetric* if $B(u,w) = B(w,u)$ for all $u,w \in V$. The set of symmetric bilinear forms on V is denoted by $V^{(2)}_{\text{sym}}$.

Exercise 1.2. Let V be a finite-dimensional vector space over F. Show that $B : \mathscr{L}(V) \times \mathscr{L}(V) \to F$ defined by

$$B(T_1, T_2) = \text{tr}(T_1 T_2)$$

is a symmetric bilinear form on $\mathscr{L}(V)$. (Recall that the trace is a similarity invariant, so $\text{tr}(T) = \text{tr}([T])$ where $[T]$ is the matrix of T with respect to any basis of V.

Definition 1.3 (Symmetric matrix)**.** An $n \times n$ matrix A is *symmetric* if $A^T = A$.

A linear operator on V may have a symmetric matrix with respect to some but not all bases of V. In contrast, the next result shows that a bilinear form on V has a symmetric matrix with respect to either all bases of V or with respect to no bases of V.

Proposition 1.4. *Suppose $B \in V^{(2)}$. Then the following are equivalent.*

(a) B is a symmetric bilinear form on V.

(b) B is a symmetric matrix for every basis of V.

(c) B is a symmetric matrix for some basis of V.

(d) B is a diagonal matrix for some basis of V.

Exercise 1.5. Prove that (a) $\implies$ (b) $\implies$ (c) $\implies$ (a) in Proposition 1.4.

We have just proved that (a), (b), and (c) of Proposition 1.4 are equivalent. Since every diagonal matrix is symmetric, (d) $\implies$ (c). To complete the proof, we will prove that (a) $\implies$ (d) by induction on $n = \dim V$.

- If $n = 1$, then (a) $\implies$ (d) because every 1×1 matrix is diagonal.
- Suppose (a) $\implies$ (d) for $\dim V$ some positive integer $k \geq 1$. If $B = 0$, then $[B] = 0$ relative to every basis of V, which is diagonal.
- Assume now that $B \neq 0$. Then there exist $u, w \in V$ such that $B(u,w) \neq 0$. Now

$$\begin{aligned} B(u+w,u+w) &= B(u,u) + B(w,w) + B(u,w) + B(w,u) \\ &= B(u,u) + B(w,w) + 2B(u,w) \quad \text{(since } B \text{ is symmetric)} \end{aligned}$$

  and therefore $2B(u,w) = B(u+w,u+w) - B(u,u) - B(w,w)$. Since the left-hand side is nonzero, the three terms on the right cannot all equal 0. Hence, there exists some $v \in V$ such that $B(v,v) \neq 0$.

  Let $U = \{u \in V : B(u,v) = 0\}$. Note that U is the kernel of the linear functional $u \mapsto B(u,v)$ on V. This linear functional is not the zero linear functional because $v \neq U$. Thus, the image is all of F, so $\dim U = \dim V - 1 = k - 1$.

- The our inductive hypothesis, there is a basis of U with respect to which the symmetric bilinear form $B|_{U \times U}$ (the restriction of B to the subspace $U \times U$) has a diagonal matrix.
- Because $v \notin U$, $(e_1, \ldots, e_{k-1}, v)$ is a linearly independent set of $k = \dim V$ vectors and hence a basis of V. Note that $B(e_i, v) = 0$ for each $i = 1, \ldots, k-1$ by definition of U. Since B is symmetric, we also have $B(v, e_i) = 0$ for all $i = 1, \ldots, k-1$. This shows that all the off-diagonal elements of the matrix of B with respect to this basis are zero, hence B is diagonal. This completes the proof that (a) $\implies$ (d), and therefore of Proposition 1.4.

2 Alternating Bilinear Forms

Definition 2.1 (Alternating bilinear form)**.** A bilinear form $B \in V^{(2)}$ is *alternating* if

$$B(u,w) = -B(w,u)$$

for all $u, w \in V$. The set of alternating bilinear forms on V is denoted by $V^{(2)}_{\text{alt}}$.

Proposition 2.2. *A bilinear form $B \in V^{(2)}$ is alternating if and only if $B(v,v) = 0$ for all $v \in V$.*

Exercise 2.3. Prove Proposition 2.2.

Exercise 2.4. Let $V = F^3$. Show that $B \in V^{(2)}$, defined by

$$B((x_1,x_2,x_3),(y_1,y_2,y_3)) = x_1y_2 - x_2y_1 + x_1y_3 - x_3y_1$$

is alternating.

Exercise 2.5. Show that, if $f, g \in V^*$, then $B \in V^{(2)}$ defined by

$$B(u,w) = f(u)g(w) - f(w)g(u)$$

is alternating.

Definition 2.6 (Antisymmetric matrix)**.** A matrix A is *antisymmetric* if $A^T = -A$.

Exercise 2.7. Show that the matrix of an alternating bilinear form on a vector space V is antisymmetric with respect to every basis of V.

Exercise 2.8. Prove that

(a) $V^{(2)}_{\text{sym}}$ and $V^{(2)}_{\text{alt}}$ are subspaces of $V^{(2)}$.

(b) $V^{(2)} = V^{(2)}_{\text{sym}} \oplus V^{(2)}_{\text{alt}}$.

Exercise 2.9. Suppose V is finite-dimensional with $\dim V = n$. Find the dimensions of $V^{(2)}_{\text{sym}}$ and $V^{(2)}_{\text{alt}}$.